$\operatorname{div}\bar{H}=0$ $\qquad$ $\operatorname{div}\bar{A}=0$

$\operatorname{rot}\bar{H}=\frac{4\pi}{c}\bar{j} \Rightarrow \Delta\bar{A}=-\frac{4\pi}{c}\bar{j} \Rightarrow \bar{A}(\bar{r})=\frac{1}{c}\int d^3\bar{r}'\,\frac{\bar{j}(\bar{r}')}{|\bar{r}-\bar{r}'|}$

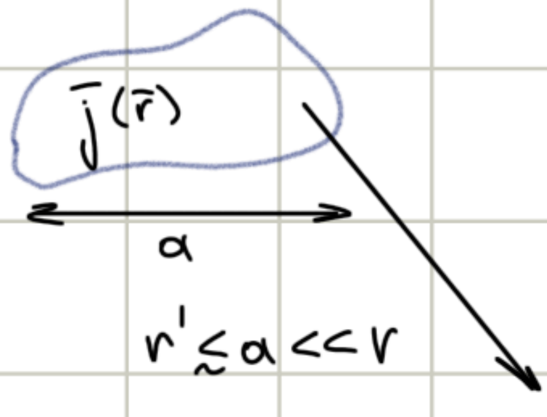
$\operatorname{div}\bar{j}=0$

$\bar{H}=\operatorname{rot}\bar{A}=\frac{1}{c}\int d^3\bar{r}'\,\frac{[\bar{j}(\bar{r}'),\,\bar{r}-\bar{r}']}{|\bar{r}-\bar{r}'|^3}$

1 Мультипольное разложение в магнитостатике

$$\frac{1}{|\bar{r}-\bar{r}'|}\simeq\frac{1}{r}+\frac{(\bar{r}\bar{r}')}{r^3}+\dots$$

«0» «1»



$$\bar{A}^{(0)}(\bar{r})=\frac{1}{cr}\int d^3\bar{r}'\,\bar{j}(\bar{r}')$$

$$0=\int d^3\bar{r}'\operatorname{div}\bar{j}\,\bar{r}'_i=\int d^3\bar{r}'\,\partial_k j_k\bar{r}'_i=\underbrace{\oint dS_k\,j_k r'_i}_{=0}-\int d^3\bar{r}'\,j_k\underbrace{\partial_k r'_i}_{\delta_{ik}}=\int d^3\bar{r}'\,j_i \rightarrow \int d^3\bar{r}'\,\bar{j}(\bar{r}')=0$$

$\Rightarrow \bar{A}^{(0)}=0$

$$\bar{A}^{(1)}(\bar{r})=\frac{1}{cr^3}\int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')$$

$$0=\int d^3\bar{r}'\operatorname{div}\bar{j}\,r_i r'_i r'_k=\int d^3\bar{r}'\,\partial_\ell j_\ell r_i r'_i r'_k=-\int d^3\bar{r}'\,j_\ell r_i\,\partial_\ell(r'_i r'_k)=-\int d^3\bar{r}'\,j_\ell r_i(\delta_{i\ell}r'_k+\delta_{\ell k}r'_i)=$$

$$=-\int d^3\bar{r}'\,(\bar{j}(\bar{r}'),\bar{r})\,r'_k-\int d^3\bar{r}'\,j_k(\bar{r}')(\bar{r}\cdot\bar{r}')$$

$$\Rightarrow \int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')=-\int d^3\bar{r}'\,(\bar{j}(\bar{r}'),\bar{r})\,\bar{r}'$$

$$[\bar{r},[\bar{r}',\bar{j}(\bar{r}')]]=\bar{r}'(\bar{r}\bar{j}(\bar{r}'))-\bar{j}(\bar{r}')(\bar{r}\bar{r}')$$

$$\int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')=-\int d^3\bar{r}'\,[\bar{r}[\bar{r}',\bar{j}(\bar{r}')]]-\int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')$$

$$\Rightarrow \int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')=\frac{1}{2}\int d^3\bar{r}'\,[[\bar{r}',\bar{j}(\bar{r}')],\bar{r}]$$

$$\bar{A}^{(1)}(\bar{r})=\frac{1}{cr^3}\int d^3\bar{r}'\,\bar{j}(\bar{r}')(\bar{r}\bar{r}')=\frac{1}{2cr^3}\left[\int d^3\bar{r}'\,[\bar{r}',\bar{j}(\bar{r}')],\bar{r}\right]=\frac{[\bar{\mu},\bar{r}]}{r^3}$$

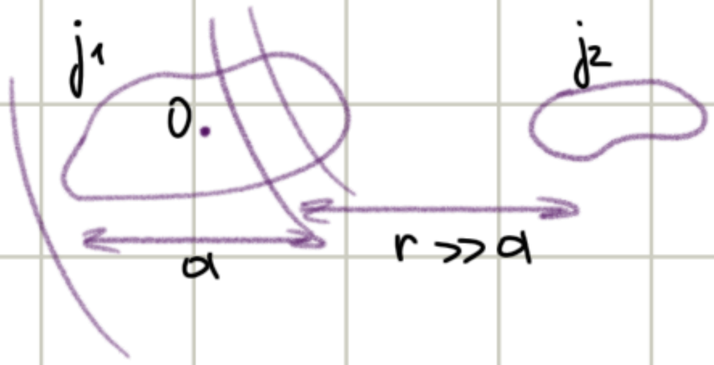
$$\bar{\mu}=\frac{1}{2c}\int d^3\bar{r}'\,[\bar{r}',\bar{j}(\bar{r}')]$$ магнитный дипольный момент

для маг. сист. зарядов: $\bar{\mu}=\sum_a\frac{e_a}{2c}[\bar{r}_a,\bar{v}_a]$

$$\bar{H}^{(1)}(\bar{r})=\operatorname{rot}\bar{A}^{(1)}(\bar{r})=\frac{3(\bar{\mu}\bar{r})\bar{r}-\bar{\mu}r^2}{r^5}=\frac{3(\bar{\mu}\bar{r})\bar{r}-\bar{\mu}}{r^3}$$ маг. поле маг. диполя

2 Сила, действ. на стат. токи со стор. маг. поля, эн-я вз-я, момент сил

$$F=\sum_a\frac{e_a}{c}[\bar{v}_a,\bar{H}(\bar{r}_a)]=\frac{1}{c}\int d^3\bar{r}'\,[\bar{j}(\bar{r}'),\bar{H}(\bar{r}')]$$



$$\bar{H}(\bar{r})\simeq\bar{H}(0)+(\bar{r}\bar{\nabla})\bar{H}(0)$$

«0» «1»

$$\bar{F}^{(0)}=\frac{1}{c}\left[\int d^3\bar{r}'\,\bar{j}(\bar{r}'),\bar{H}(0)\right]=0$$

$$\bar{F}^{(1)}=\frac{1}{c}\int d^3\bar{r}'\,[\bar{j}(\bar{r}'),(\bar{r}\bar{\nabla})\bar{H}(0)]\ *$$

$(*)_i = \varepsilon_{ijk} j_j(\bar r') r'_e \nabla_e H_k(0) = \varepsilon_{ijk} j_j(\bar r') r'_e \nabla^{(\bar r)}_k H_e(0) = \varepsilon_{ijk} j_j(\bar r') r'_e \frac{\partial}{\partial r_k} H_e(\bar r)\Big|_{\bar r=0} =$

$\operatorname{rot}\bar H = \frac{4\pi}{c}\bar j_2 = 0$ т.к. j_2 нет в этой обл.

$\operatorname{rot}\bar H = 0 \Rightarrow \nabla_e H_k = \nabla_k H_e$

$= -\varepsilon_{ikj}\frac{\partial}{\partial r_k}\left(j_j(\bar r') r'_e H_e(\bar r)\right)\Big|_{\bar r=0} = -\operatorname{rot}_{\bar r}\left(\bar j(\bar r')(\bar r' \bar H(\bar r))\right)\Big|_{\bar r=0}$

$\bar F^{(1)} = -\operatorname{rot}_{\bar r}\left(\frac{1}{c}\int d^3\bar r'\, \bar j(\bar r')(\bar r', \bar H(\bar r))\right)\Big|_{\bar r=0} = -\operatorname{rot}_{\bar r}[\bar\mu, \bar H(\bar r)]\Big|_{\bar r=0}$

$(\operatorname{rot}[\bar\mu, \bar H(\bar r)])_i = \varepsilon_{ijk}\partial_j \varepsilon_{klm}\mu_e H_m = (\delta_{ie}\delta_{jm} - \delta_{im}\delta_{je})\mu_e \partial_j H_m = \mu_i \underbrace{\operatorname{div}\bar H}_{0} - (\bar\mu \cdot \bar\nabla) H_i = -(\bar\mu, \bar\nabla) H_i$

$\bar F^{(1)} = (\bar\mu\bar\nabla)\bar H(0)$

$0 = [\bar\mu, \operatorname{rot}\bar H]_i = \varepsilon_{ijk}\mu_j \varepsilon_{klm}\partial_e H_m = (\delta_{ie}\delta_{jm} - \delta_{im}\delta_{je})\mu_j \partial_e H_m = \mu_j \partial_i H_j -$

$-(\bar\mu\bar\nabla) H_i = \partial_i(\bar\mu\bar H) - (\bar\mu\bar\nabla) H_i$

$\bar F^{(1)} = \bar\nabla(\bar\mu\bar H) = -\bar\nabla(-\bar\mu\bar H)$

$\Rightarrow U = -(\bar\mu\bar H)$ энергия взаимод. маг. диполя с маг. полем

Момент сил

$\bar M = \sum_a [\bar r_a, \bar F_a] = \sum_a [\bar r_a, \frac{e_a}{c}[\bar v_a, \bar H(\bar r_a)]] = \frac{1}{c}\int d^3\bar r' [\bar r', [\bar j(\bar r'), \bar H(\bar r')]]$

$\bar j(\bar r') = \sum_a e_a \bar v_a \delta(\bar r - \bar r_a)$

$\bar H \simeq \bar H(0) = \text{const}$

$\bar M^{(0)} = \frac{1}{c}\int d^3\bar r' [\bar r', [\bar j(\bar r')\bar H(0)]] = \frac{1}{c}\int d^3\bar r' \bar j(\bar r')(\bar r', \bar H(0)) - \frac{1}{c}\bar H(0)\underbrace{\int d^3\bar r' (\bar r', \bar j(\bar r'))}_{(*)} =$

$= [\bar\mu, \bar H]$

$\operatorname{div}(r'^2\bar j) - \operatorname{div}\bar j$

$(\bar r', \bar j(\bar r'))$

$\operatorname{div}(r'^2\bar j(\bar r')) = r'^2 \operatorname{div}\bar j + 2(\bar r', \bar j(\bar r')) = 2(\bar r', \bar j(\bar r'))$

$(*) = \frac{1}{2}\int d^3\bar r' \operatorname{div}(r'^2 \bar j(\bar r')) = 0$

$\bar M^{(0)} = [\bar\mu, \bar H(0)]$

$\frac{d\bar L}{dt} = \bar M$

$\bar L = \sum_a [\bar r_a, \bar p_a] \simeq \sum_a m_a [\bar r_a, \bar v_a] = c\sum_a \frac{m_a}{e_a}\frac{e_a}{c}[\bar r_a, \bar v_a] = \frac{2cm}{e}\sum_a \frac{e_a}{2c}[\bar r_a, \bar v_a] = \frac{2mc}{e}\bar\mu$

$\frac{v_a}{c} \ll 1$, $\bar p_a \simeq m_a \bar v_a$, $\frac{e_a}{m_a} = \frac{e}{m} = \text{const}$

$\bar\mu = \frac{e}{2mc}\bar L$ гиромагнитное соотношение

$(\bar\mu = g\frac{e}{2mc}\bar S)$

$\frac{d\bar L}{dt} = [\bar\mu, \bar H] = \frac{e}{2mc}[\bar L, \bar H] = [\bar\Omega, \bar L]$, $\bar\Omega = -\frac{e\bar H}{2mc}$

ларморовская частота



3 Плоские электромагнитные волны

$\Box A^\mu = -\frac{4\pi}{c} j^\mu$, лор. калибр. $\partial_\mu A^\mu = 0$ $\left(\frac{1}{c}\frac{\partial \varphi}{\partial t} + \operatorname{div}\bar{A}\right) = 0$

$\Box = \partial_\nu \partial^\nu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta$

Пустое пр-во $j^\mu = 0$ $\partial_\nu \partial^\nu A^\mu = 0$

$\partial_\mu A^\mu = 0$

Ищем реш. в виде $A^\mu = A^\mu_0 \cos(k_\nu r^\nu - \varphi) = A^\mu_0 \cos(\underbrace{\omega t - \bar{k}\bar{r}}_{\text{скаляр}} - \underbrace{\varphi}_{\text{скаляр}}) = \operatorname{Re}\left[\underbrace{A^\mu_0 e^{-i\varphi}}_{a^\mu_0 \text{ пост. комплекс. 4-в-р}} e^{ik_\nu r^\nu}\right]$

k^ν - пока произв. 4-в-р $= \left(\frac{\omega}{c}, \bar{k}\right)$

$\partial_\nu r^\mu = \delta^\mu_\nu$ | $\partial_\nu (e^{ik_\alpha r^\alpha}) = e^{ik_\alpha r^\alpha} ik_\alpha \partial_\nu r^\alpha = ik_\nu e^{ik_\alpha r^\alpha}$

$\partial^\nu r_\mu = \delta^\nu_\mu$ | $\partial^\nu (e^{ik^\alpha r_\alpha}) = ik^\nu e^{ik^\alpha r_\alpha}$

$\partial_\nu \partial^\nu A^\mu = \partial_\nu \partial^\nu (a^\mu_0 e^{ik_\alpha r^\alpha}) = a^\mu_0 (ik^\nu)(ik_\nu) e^{ik_\alpha r^\alpha} = -a^\mu_0 k_\nu k^\nu e^{ik_\alpha r^\alpha} = 0$

$\Rightarrow k_\nu k^\nu = 0 \Rightarrow \left(\frac{\omega}{c}\right)^2 - \bar{k}^2 = 0$ дисперс. соотн. $(\omega = \pm c|\bar{k}|)$

$\partial_\mu A^\mu = ik_\mu a^\mu_0 e^{ik_\alpha r^\alpha} = 0 \Rightarrow k_\mu a^\mu_0 = 0$ лор. инв. форм.